

Danyal Gu

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EDUCATION

University of Waterloo

Candidate for B.ASc. in Honours Mechanical Engineering

Waterloo, ON

Sep 2025 - Apr 2030

- **Relevant Courses:** Mechanics of Deformable Solids, Control of Material Properties, Electromechanical Devices
- **Scholarships:** President's Scholarship of Distinction

EXPERIENCE

Undergraduate Research Assistant

University of Waterloo (Multi-Physics Interaction Lab)

Sep 2026 - Present

Waterloo, ON

- Selected as a URA through the President's Research Award to support Prof. Jean-Pierre Hickey's (MPI Lab) rotating detonation engine project, targeting 1,000 lbf thrust as the second Canadian university-led RDE program
- Assigned to contribute to research, design, and analysis funded through Launch Canada, beginning with a literature review of RDE research ahead of CFD/HPC work in OpenFOAM

Mechanical Design Co-op

TK Modular Systems

Apr 2026 - Aug 2026

Oakville, ON

- Modified 3D CAD assemblies in PTC Creo for two large-scale biopharmaceutical water purification skid systems (40+ skids each), interpreting P&IDs to implement client-requested structural and piping design changes
- Created 150+ new and revised 300+ existing fabrication, spool, conduit, and structural frame drawings spanning 4,000+ components, applying GD&T to produce actionable shop-floor documentation
- Managed the drawing release lifecycle in Windchill PDM, resolving cross-functional markups from mechanical and process engineering reviews against client specifications and internal design standards

Chassis Design & Analysis Engineer

University of Waterloo Formula SAE (FSAE)

Sep 2025 - Present

Waterloo, ON

- Designed a brake line mounting system using a standardized fixture deployed at 20 points along the carbon fiber floor, resolving critical packaging constraints to ensure safe line retention
- Developed a 3D-printed welding fixture using in-context assembly references, creating a mechanical hard-stop to position the master switch below the helmet impact zone per FSAE safety rules, achieving <1mm vertical accuracy and cutting setup time to 5 minutes with zero post-weld adjustments
- Machined interference-fit assemblies and structural inserts from aluminum and steel stock, utilizing datum alignment on manual mills and lathes to hold strict dimensional tolerances within $\pm 0.01\text{mm}$

PROJECTS

FSAE 2026 Chassis Torsional Stiffness Analysis

ANSYS Static Structural, SolidWorks, Excel

- Built a finite element beam spaceframe model in ANSYS Static Structural using spherical joints to replicate suspension kinematics, adding accumulator and AIP geometry to evaluate their structural contribution
- Evaluated four frame configurations — baseline, with AIP, with accumulator, and combined — identifying that the accumulator adds 25.6% torsional stiffness and the full configuration achieves 2660 Nm/deg, a 31% improvement over the baseline frame
- Designed a rigid test fixture to interface the chassis with the physical torsion rig, enabling future experimental correlation of the FEA model to confirm the frame meets handling targets

Gantry-Driven Checker Robot

SolidWorks, 3D Printing, C++

- Designed the complete mechanical assembly in SolidWorks - gravity-fed hopper, rotating gate dispenser, and 3-axis rack and pinion gantry system - and led full physical assembly of the prototype
- Selected filament materials based on mechanical requirements, PETG for structural racks, carbon fiber for high-stress pinions, PLA for tolerance-critical sensor enclosures
- Achieved 94% colour detection accuracy and 90% mechanical reliability across 10 trials; identified gantry frame deflection as the primary source of 75% placement accuracy, proposing guide rod additions as the structural fix

SKILLS

Mechanical Design & Simulation: PTC Creo, SolidWorks, FEA (ANSYS), CFD (Pointwise), DFMA

Drafting & PDM: AutoCAD, GD&T, PTC Windchill PDM

Technical Tools: MATLAB, C++, Java, Python, Microcontrollers, 3D Printing